

Introduction to Linear Algebra Using Matlab

Participated in & did the coursework [Y/N]?

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1 INTRODUCTION

1.1 Abstract

This report summarizes Laboratory Exercise 2, which focuses on the time-domain representation of signals and their corresponding operations. The primary objectives include determining symbolic mathematical expressions for continuous-time signals and utilizing MATLAB to sketch these sequences accurately. Essential signal operations such as time shifting, time reversal, and time compression are applied to a base signal to visually demonstrate their theoretical effects in the time domain. Furthermore, the coursework explores the mathematical formulation and plotting of impulse responses for standard filters, specifically low-pass, high-pass, bandpass, and bandstop filters. Finally, the auto-correlation of a specified signal is computed and graphed against its lag time to analyze its self-similarity.

1.2 Conspectus

1.3 What are the objectives of the coursework?

- 1) To write symbolic math expressions for continuous-time signals and specific filter impulse responses;
- 2) To successfully sketch multiple sequences demonstrating time-domain operations like shifting, flipping, and compression;

1.4 How does the coursework fit the course and previously done coursework?

By:

- 1) Applying practical MATLAB programming to execute theoretical signal operations covered in the Signals and Systems course.
- 2) Translating symbolic math expressions into visual time-domain representations for deeper behavioral analysis.

By affixing my/our signature/s, I/we, the author/s, pledge that: I/we have completed this coursework on my/our own; I/we have not used any unauthorized material/assistance/help on this coursework; and I/we have not given directly or indirectly to any other student/unauthorized person/means any access to any part of the specified coursework. Coram Deo.

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1.5 How were the objectives achieved?

By:

- 1) Utilizing MATLAB functions such as heaviside, dirac, and exp to symbolically define the input signals and filter responses.
- 2) Using plotting commands like subplot to arrange multiple graphs into a single figure and xcorr to determine the auto-correlation.

1.6 What are the key results and generalizations?

The key results are:

- 1) Time-domain operations were visually verified, showing accurate signal shifts, reversals, and scaling based on their mathematical adjustments.
- 2) The impulse responses for various filters were successfully plotted, and the auto-correlation effectively displayed the relationship between the signal's amplitude and lag τ .

1.7 Principles

1.8 What are the necessary and relevant concepts, principles, theoretical and design considerations for understanding the coursework and for supporting the correct results?

- 1) Signal operations modify a signal's independent variable, involving transformations like time shifting $x(t-2)$, time reversal $x(-t)$, and time compression $x(2t)$.
- 2) Impulse responses define the output behavior of linear time-invariant systems, such as low-pass filters characterized by $h_1(t) = e^{-10t}u(t)$ and bandpass filters utilizing a cosine carrier.

1.9 How does any new component, not covered in previous coursework, function?

By:

- 1) Utilizing the heaviside(t) function to act as a unit step function $u(t)$, which bounds signals mathematically within specific time intervals.
- 2) Applying the xcorr function to compute the cross-correlation sequence, which is then scaled by the time differential to correctly evaluate auto-correlation.

1.10 What figures, equations, and/or tables could support your answers in Sec. ?? and Sec. ???

- Figure 1 supports the signal operations concepts by displaying $x(t)$ alongside its shifted, flipped, and compressed variants across six structured subplots.
- Figure 4 supports the auto-correlation evaluation by plotting the resulting amplitude $R_{xx}(\tau)$ against the lag τ in seconds.

1.11 Did you cite more than two publications in your answers in Sec. ?? and Sec. ???

Yes.

1.12 Did you cite any online source in your answers in Sec. ?? and Sec. ???

No.

2 EXPERIMENTAL DESIGN

2.1 How does your implementation in Sec. ?? achieve the objectives?

By:

- Formulating four simultaneous mesh equations based on the circuit diagram loops.
- Converting the linear system into matrix form $A\mathbf{i} = \mathbf{c}$ for computational processing.

2.2 Why does your implementation in Sec. ?? achieve the objectives?

Because:

- Matrix representation enables the application of algorithmic solutions like Cramer's Rule and Matrix Inverse.
- The Symbolic Math Toolbox allows for exact algebraic manipulation of variables before numerical evaluation.

2.3 How does your evaluation in Sec. ?? achieve the objectives?

By:

- Solving the system using three distinct methods: Cramer's Rule, Matrix Inverse, and the built-in 'solve' function.
- Computing eigenvalues and eigenvectors using both characteristic polynomials and the standard 'eig' function.

2.4 Why does your evaluation in Sec. ?? achieve the objectives?

Because:

- Comparing results across multiple calculation methods ensures the numerical accuracy of the calculated currents.
- Verifying manual algorithmic logic (loops/determinants) against optimized built-in functions validates the code structure.

2.5 Implementation

Rule of thumb: Implementation is how you made your work; (keywords: implemented, created, made, soldered, programmed, etc.).

2.5.1 What were the materials used?

- Personal Computer with MATLAB software installed.
- Laboratory exercise manual and circuit diagram for mesh analysis.

2.5.2 What is the summary of the processes used to make the coursework?

The implementation involved defining the circuit equations symbolically and creating a script to solve the linear algebra problem. A pseudocode for the general solving process is shown in Table ??.

- Defined symbolic variables i_1, i_2, i_3, i_4 and wrote mesh equations based on KVL.
- Constructed the coefficient matrix A and constant vector $\mathbf{c}$ from the linear equations.

2.6 Evaluation

Rule of thumb: Evaluation is how you tested your work for correctness; (keywords: measured, tested, compared, simulated, etc.).

2.6.1 What were your procedures for evaluating the correct outcome of your coursework?

- The calculated currents from Cramer's Rule were cross-referenced with the Matrix Inverse method results.
- Eigenvalues derived from the 'poly' and 'roots' functions were checked against the 'eig' function output.

2.6.2 What quantities were gathered and how have you obtained them for testing the veracity of your results?

- Loop currents ($i_1 \approx 0.0556$, $i_2 \approx 0.0338$, etc.) obtained via determinants and inverses.
- Eigenvalues (e.g., 141.7814) and Eigenvectors were gathered to analyze the system matrix properties.

Table 1
Pseudocode for Solving System of Linear Equations

Input(s):	
A	: Coefficient Matrix (4×4)
$\mathbf{c}$	: Constant Vector (4×1)
Output(s):	
$\mathbf{i}$	: Loop Currents vector (4×1)

Require: $\det(A) \neq 0$

Ensure: $A \cdot \mathbf{i} = \mathbf{c}$

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1: Define symbolic variables eqn1 ... eqn4
2:  $A \leftarrow$  Extract Coefficients from equations
3:  $\mathbf{c} \leftarrow$  Extract Constants from equations
4: Calculate Determinant  $D \leftarrow \det(A)$ 
5: if Method is Cramer's Rule then
6:   for  $k = 1$  to  $4$  do
7:      $A_k \leftarrow A$ 
8:     Replace  $k$ -th column of  $A_k$  with  $\mathbf{c}$ 
9:      $i_k \leftarrow \det(A_k) / D$ 
10:  end for
11: else if Method is Matrix Inverse then
12:   $A_{inv} \leftarrow \text{inv}(A)$ 
13:   $\mathbf{i} \leftarrow A_{inv} \times \mathbf{c}$ 
14: end if
15: Display  $\mathbf{i}$ 
```

3 ANALYSIS OF RESULTS

3.1 How do the results achieve the objectives?

By:

- 1) Calculating the specific mesh currents i_1 through i_4 using Cramer's Rule, Matrix Inverse, and the `solve` function.
- 2) Determining the characteristic polynomial, eigenvalues, and eigenvectors of the circuit's coefficient matrix.

3.2 Why do the results achieve the objectives?

Because:

- 1) The consistency of current values across three different calculation methods verifies the accuracy of the linear algebra implementation.
- 2) The extraction of eigenvalues demonstrates the successful application of spectral theory to the circuit matrix using MATLAB.

3.3 Are all your results correct in accordance with what you described in Sec. ?? evaluation process? Why?

Yes, because:

- 1) The output for current i_1 remained exactly 0.0556 A regardless of whether Cramer's rule or Matrix Inverse was used.
- 2) The eigenvalues calculated manually via the characteristic polynomial matched the outputs of the built-in `eig(A)` function.

3.4 What is Result 1 (Mesh Currents), what does it mean if it is correct, and how does it contribute to reaching the objectives?

- 1) Table ?? (or the MATLAB output in the Appendix) displays the calculated loop currents: $i_1 = 0.0556$, $i_2 = 0.0338$, $i_3 = 0.0717$, and $i_4 = 0.1009$.
- 2) These positive values indicate that the actual current flow direction matches the assumed clockwise direction of the mesh loops.
- 3) There were no discrepancies; the values satisfied the original KVL equations when substituted back.
 - a) The Matrix Inverse method returned identical floating-point values to the symbolic solution.
 - b) Cramer's rule determinants yielded the same ratios.
- 4) Standard circuit analysis textbooks confirm that unique solutions exist for linear resistive circuits with independent sources [1].
- 5) This result confirms that the system of linear equations $Ai = c$ was correctly formulated and solved.

3.5 What is Result 2 (Eigenvalues), what does it mean if it is correct, and how does it contribute to reaching the objectives?

- 1) The eigenvalues of Matrix A were found to be $\lambda \approx 141.78, 115.27, -66.49, -146.56$.
- 2) These values represent the scalar factors by which the eigenvectors are scaled by the linear transformation of the circuit matrix.
- 3) Discrepancies were negligible, limited only to minor floating-point variations in the 4th decimal place.

- a) The characteristic polynomial calculation `roots(p)` aligned with the direct `eig(A)` function.
 - b) Small numerical noise (e.g., -0.0000) in the polynomial vector did not impact the final roots significantly.
- 4) This analysis is standard in linear algebra coursework for understanding matrix properties [2].
 - 5) This result validates the student's ability to manipulate matrices beyond simple system solving, covering spectral concepts.

3.6 Did you cite more than two publications in your answers above (yes/no)?

No.

4 DISCUSSION AND CONCLUSION

4.1 What are the main points that should be known, remembered, and learned about the coursework?

- 1) Linear algebra provides a systematic method for solving complex circuit problems by converting Mesh Analysis equations into the matrix form $A \cdot i = c$.
- 2) All implemented calculation methods (Cramer's Rule, Matrix Inverse, and MATLAB's `solve`) yielded identical results, confirming that the choice of mathematical approach does not alter the physical solution of the circuit.

4.2 What are the gists of the inferences drawn from your results?

- 1) The calculated mesh currents ($i_1 = 0.0556$ A, $i_2 = 0.0338$ A, $i_3 = 0.0717$ A, $i_4 = 0.1009$ A) confirm that the system is consistent and the circuit has a unique solution.
- 2) The successful extraction of eigenvalues ($\lambda \approx 141.78, 115.27, -66.49, -146.56$) infers that the Symbolic Math Toolbox is effective for analyzing the spectral properties of circuit matrices, which is essential for understanding system stability in advanced applications.

4.3 Briefly, what are your comments on (1) your results, and (2) future coursework if any?

- 1) The results are highly accurate with negligible discrepancies, as the manual symbolic derivation perfectly matched the numerical functions of MATLAB.
- 2) For future coursework, these linear algebra techniques should be applied to AC circuits with complex impedances to test the robustness of the matrix methods against complex numbers.

5 AUTHORS CONTRIBUTION

On what contributions to specify, see the terms at www.elsevier.com.

- 1) BAUTISTA, Alyssa Nicole : Methodology, Results.
- 2) GONZALES, Jeremia E.: Abstract, Conspectus, Principles.
- 3) MENDIOLA, Keifer Judd: Conclusions.

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- [1] J. W. Nilsson and S. A. Riedel, Electric Circuits, 11th ed. New York, NY, USA: Pearson, 2018. [2] G. Strang, Linear Algebra and Its Applications, 5th ed. Wellesley, MA, USA: Wellesley-Cambridge Press, 2016.